

Duality

Duality in linear optimization

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Optimization: principles and algorithms



Duality in linear optimization

We consider now the specific case of linear optimization.

We show that the dual problem has some nice properties in this context.

Linear optimization

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$Ax = b$$

$$x \geq 0$$

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$c \in \mathbb{R}^n$$

p. 102

$$h(x) = b - Ax, g(x) = -x$$

$$L(x, \lambda, \mu) = c^T x + \lambda^T (b - Ax) - \mu^T x = (c - A^T \lambda - \mu)^T x + \lambda^T b$$

Linear optimization

$$\min_{x \in \mathbb{R}^m} -b^T x$$

subject to

p. 103

$$A^T x \leq c$$

$$g(x) = A^T x - c$$

$$L(x, \mu) = -b^T x + \mu^T (A^T x - c) = (-b + A\mu)^T x - \mu^T c$$

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$c \in \mathbb{R}^n$$

Dual problem

Theorem 4.14

The dual of a linear optimization problem is another linear optimization problem.
The dual can be derived for each possible specification.

Theorem 4.15

The dual of the dual is the primal.

Strong duality

Theorem 4.17

- Consider the primal problem and its dual.

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- ▶ If one problem has an optimal solution, so does the other one,

Strong duality

Theorem 4.17

- ▶ Consider the primal problem and its dual.
- ▶ If one problem has an optimal solution, so does the other one,
- ▶ and the optimal value of their objective functions are the same.

Summary

The dual of a linear optimization problem happens to be also a linear optimization problem.

The dual of the dual is the primal.

The strong duality theorems states that the best lower bound is tight.